



# HALDIA INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

**Paper Code: PCC-CS 503**

**Paper Name: Machine Learning**

*Time Allotted: 3 Hours*

*Full Marks: 70*

*The figures in the margin indicate full marks*

*Candidates are required to give their answers in their own words as far as practicable*

## Group – A

### (Multiple Choice Type Questions)

**15 x 1 = 15**

**Choose the correct alternatives from the followings:**

1. (i) A software designed to evaluate whether a customer account is hacked or not is an example of
  - a) Unsupervised learning
  - b) supervised classification
  - c) Supervised regression
  - d) None of these
- (ii) How many coefficients do you need to estimate a simple regression model (one independent variable)?
  - a) 1
  - b) 2
  - c) 3
  - d) 4
- (iii) Which of the following is a measure of the spread of data?
  - a) Mean
  - b) Variance
  - c) Covariance
  - d) All of the above
- (iv) Which of the following is an alternative technique to principal component analysis?
  - a) Factor analysis
  - b) Independent component analysis
  - c) Latent semantic analysis
  - d) All of the above
- (v) Which supervised learning technique can process both numeric and categorical input attributes?
  - a) Bayes classifier
  - b) Linear regression
  - c) Logistic regression
  - d) None of the above
- (vi) Naïve Bayes Algorithm belongs to which category?
  - a) Supervised
  - b) Reinforcement
  - c) Unsupervised
  - d) None of these
- (vii) \_\_\_ is the machine learning algorithm that can be used with unlabeled data.
  - a) Regression algorithm
  - b) Clustering algorithm
  - c) Instance-based algorithm
  - d) All of the above
- (viii) What does K stand for in K mean algorithm?
  - a) Number of clusters
  - b) Number of data
  - c) Number of attributes
  - d) Number of iterations
- (ix) Boosting builds ensembles in \_\_\_\_\_.
  - a) Series
  - b) Parallel
  - c) Series and parallel
  - d) None of the above
- (x) Bagging comes under which \_\_\_\_\_ method?
  - a) Sequential methods
  - b) Parallel method
  - c) Line method
  - d) Dot method
- (xi) An orderly set of data arranged in accordance with their time of occurrence is called:
  - a) Arithmetic series
  - b) Harmonic series
  - c) Geometric series
  - d) Time series
- (xii) Secular trend can be measured by:
  - a) Two methods
  - b) Three methods
  - c) Four methods
  - d) Five methods
- (xiii) The unsystematic sequence which follows irregular pattern of variations is called:
  - a) Noise
  - b) Signal
  - c) Linear
  - d) Non-linear

(xiv) The general limitations of back propagation rule is/are

- a) Scaling      b) Slow convergence      c) Local minima problem      d) All of the above

(xv) How many types of feedback does reinforcement provide?

- a) 1      b) 2      c) 3      d) 4

**Group – B**

**(Short Answer Type Questions)**

**Attempt any three from the followings:**

**3 x 5 = 15**

2. How logistic regression is different compared to linear regression? Explain with suitable example.

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3. (i) What is the Euclidian distance between two data points: A(1,3) and B(2,3)?

(ii) Which machine learning algorithm is called lazy learner and why?

2+3

4. (i) What do you mean by clustering?

(ii) Write note on the different types of clustering.

1+4

5. (i) Define seasonal index.

(ii) Explain cyclic variations.

2+3

6. (i) State the need of scaling in machine learning.

(ii) Name some scaling techniques in ML with example

2+3

**Group – C**

**(Long Answer Type Questions)**

**Attempt any four from the followings:**

**4 x 10 = 40**

7. (i) The sales of a company (in million dollars) for each year are shown in the table bellow

| S. No. | Year | Sales |
|--------|------|-------|
| 1      | 2005 | 12    |
| 2      | 2006 | 19    |
| 3      | 2007 | 29    |
| 4      | 2008 | 37    |
| 5      | 2009 | 45    |

Find the least square regression line,  $y = ax + b$ .

(ii) Use the least square regression line as a model to estimate the sales of the company in 2012.

5+5

8. (i) Explain matrix factorization and where it is used?

(ii) What are the advantages and disadvantages of PCA algorithm?

5+5

9. (i) What is the general idea of boosting technique?

(ii) Compare bagging and boosting.

5+5

10. (i) Explain the method of fitting a straight line.

(ii) State the two normal equations used in fitting a straight line.

5+5

11. (i) Applying the k-means algorithm, find two clusters in the following data.

| $x$ | 185 | 170 | 168 | 179 | 182 | 188 |
|-----|-----|-----|-----|-----|-----|-----|
| $y$ | 72  | 56  | 60  | 68  | 72  | 77  |

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12. (i) Explain the architecture of an artificial neural network (ANN).

(ii) What is the role of the Activation functions in Neural Networks?

5+5